

NIHAO NIU

acmachineoier@gmail.com | Anhui University, Hefei, China

EDUCATION

Anhui University (AHU) — School of Artificial Intelligence

Sep. 2024 – Present

B.Eng. in Artificial Intelligence | GPA: 3.8 / 5.0 (88%)

National Double First-Class University (211 Project)

National University of Singapore (NUS) — Short-term Study Visit

Feb. 2026

Short-term Study Visit, instructed by Dr. Sanka Rasnayaka

RESEARCH INTERESTS

Multimodal Large Language Models · Visual Token Compression · Efficient Deep Learning · Neural Network Pruning

PUBLICATIONS

[1] **Nihao Niu**, Jingyuan Zhang, Ke Xu. *Register-Guided Context Scoring for Visual Token Compression in Multimodal LLMs*. In ACM International Conference on Multimedia (ACM MM 2026), Under Review. **[First Author]**

RESEARCH EXPERIENCE

Visual Token Compression for Multimodal LLMs

Jun. 2025 – Present

Advised by Prof. Ke Xu | Anhui University

- Proposed Register-Guided Context Scoring, a shared-attention dual-path scorer (13.8M params) for visual token compression in Multimodal LLMs, achieving 2.23× prefill speedup at 5% token retention on Qwen2.5-VL-7B with backbone frozen throughout training.
- Designed a dual-path scorer with learnable register tokens as global semantic anchors, adaptive context gating, and bounded proportional modulation; registers self-organize into functionally distinct roles (content-adaptive activator, global suppressor, semantic modulator, calibration anchor) without explicit supervision.
- Evaluated on 8 multimodal benchmarks (POPE, MME, OCRBench, DocVQA, TextVQA, AI2D, MMMU, ScienceQA) via LMMs-Eval; at 5% retention outperformed the strongest learnable baseline by +2.5pp average retention, +158.7 MME, and +2.98 AI2D, and surpassed the best training-free method by +233 MME and +18.7 DocVQA ANLS.
- Demonstrated zero-shot generalization to video understanding (MVBench, +0.57 at 10% retention) without any video-specific training; submitted as first author to ACM MM 2026, currently under review.

TECHNICAL SKILLS

Languages: Python, C++
Frameworks & Libraries: PyTorch, Keras, scikit-learn, Hugging Face Transformers
Research Tools: LMMS-Eval, CLIP/ViT architectures, multimodal benchmarks (MME, MMBench, SeedBench)

HONORS & AWARDS

Academic Excellence Scholarship (学习优秀奖) — Anhui University	2024 – 2025
CSP-J (Junior) — First Prize (一等奖)	2018, 2019, 2020
CSP-S (Senior) — Second Prize (二等奖)	2019, 2021

LANGUAGES

Chinese (Native) | English (Professional Working Proficiency)